Capstone Project 2024-26

Final Report

Title of the Project

Machine Learning-Based Heart Attack Prediction: A Logistic Regression Approach

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CERTIFICATE FROM THE FACULTY GUIDE

This is to certify that the Project Work titled **Machine Learning Based-Heart Attack Risk Prediction: A Logistic Regression Approach** is a Bonafide work carried out by **Ms. Vaishali** , Roll No. **021330424044,** an MBA student. 2024-2026 of ITM Business School, Kharghar, Navi Mumbai, under my supervision & guidance.

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Vaishali

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ABSTRACT

Heart attacks continue to be one of the most critical health challenges globally, accounting for nearly one-third of all deaths each year. The growing burden of cardiovascular diseases has led to increased interest in early risk identification systems. Existing research highlights that many heart attack cases are linked to preventable or modifiable risk factors such as cholesterol, blood pressure, smoking, diabetes, obesity, and lifestyle pressures. Recent studies have shown that machine learning techniques can significantly improve the prediction of cardiac events by analysing complex interactions among clinical and behavioural variables Building on this foundation, the present study develops a data-driven model to assess heart attack risk using a dataset of 50,000 patient health records. The dataset captures a wide range of parameters, including demographic details, lifestyle behaviours, medical histories, and real-time health indicators such as cholesterol levels, BMI, blood pressure, glucose, and heart rate. Extensive preprocessing was carried out to improve data quality—missing values were treated, categorical variables were encoded, numerical features were normalised, and outliers were reviewed to improve model reliability.

Multiple machine learning algorithms were tested, including Logistic Regression, Random Forest, and XGBoost. Among them, Logistic Regression performed notably well with an accuracy of 86.69%, offering a simple yet dependable approach for binary classification of heart attack risk. These results align with earlier studies where supervised ML models, especially ensemble methods, have demonstrated strong predictive performance in cardiovascular risk assessment The findings of this project highlight several significant predictors of heart attack risk—particularly diabetes, high cholesterol, elevated BMI, hypertension, and persistent chest pain. Behavioural patterns such as smoking, low physical activity, and stress levels also showed clear associations with increased risk. By integrating clinical factors with lifestyle data, the model provides a more holistic risk assessment that is practical for real-world healthcare scenarios.

Overall, the study reinforces the potential of machine learning as a supportive tool for healthcare professionals. Such systems can assist in early screening, personalised medical planning, and preventive health strategies. However, the work remains limited by the use of secondary data and lack of real-time clinical validation. Future extensions may include deep learning models, wearable device integration, and collaboration with medical institutions to further enhance accuracy and clinical applicability.

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CHAPTER- 1

INTRODUCTION

1.1 General Overview of the Sector

The healthcare industry plays a crucial role in patient care, diagnosis, treatment, and disease prevention. Over the years, advancements in medical science and technology have transformed healthcare, enabling predictive care and personalized treatments.

One of the most pressing concerns in healthcare today is cardiovascular diseases (CVDs), particularly heart attacks, which are a leading cause of death worldwide. According to the World Health Organization (WHO), CVDs are responsible for 17.9 million deaths each year, making up 31% of global fatalities. Of these, around 85% result from strokes and heart attacks.

Over 17.9 million people die each year from heart attacks, strokes, and other cardiovascular illnesses (CVDs), which cause about 31% of all fatalities worldwide. These disorders develop when a constricted or obstructed blood artery restricts blood flow to the heart or brain. Diseases affecting the heart's muscles, valves, and rhythm are also considered heart disease. These illnesses are sometimes called "silent killers," and early diagnosis might be difficult since they may not be discovered until a serious or deadly incident happens. Preventive care and early detection are therefore essential for lowering death rates.

Although they provide useful information, traditional diagnostic techniques including electrocardiograms, blood pressure checks, cholesterol testing, and blood sugar evaluations have drawbacks. These methods may be expensive, time-consuming, and may postpone essential medical procedures, which could put patients at greater danger.

Artificial intelligence (AI) and machine learning (ML) have been increasingly popular in the medical industry in recent years. These technologies make predictive analytics, risk assessment, and automated decision-making possible. More precise diagnoses, individualized treatment plans, and better medical decision-making are all made possible by AI and ML, which allow healthcare practitioners to quickly handle enormous volumes of data.

The need for sophisticated diagnostic tools, such as predictive models driven by artificial intelligence (AI) and machine learning (ML), has increased due to the rising incidence of cardiovascular disorders. By evaluating patient data to find risk variables, these models facilitate early detection and the use of preventive measures.

1. Machine Learning Approaches in Healthcare

For addressing complex healthcare challenges, including heart attack risk prediction, different Machine Learning approaches are utilized. These include:

Supervised Learning: Involves training models on labeled datasets where both input features and output labels are known. This approach is used in disease prediction, patient classification, and medical image recognition. Common algorithms include Decision Trees, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Random Fores

Unsupervised Learning: Deals with unlabeled datasets, where the algorithm identifies patterns and relationships without predefined categories. Clustering and association rule learning are the primary techniques used in medical anomaly detection, patient segmentation, and genetic data analysis. Algorithms like K-Means Clustering and Hierarchical Clustering are widely applied.

Semi-supervised learning: A combination of supervised and unsupervised learning, where models are trained on labeled and unlabeled data. This approach is commonly used in medical research, disease classification, and speech recognition.

Reinforcement Learning: Focuses on interaction with an environment where an agent learns through trial and error, receiving rewards for correct actions. It is applied in robotic surgeries, automated diagnostics, and AI-driven healthcare management systems.

2. Machine Learning Classification Techniques

In the domain of heart attack risk prediction, ML classification techniques play a crucial role in categorizing patients into different risk levels based on medical records, genetic factors, and lifestyle habits. Some widely used ML techniques include:

Regression Models (Linear & Logistic Regression): Used for predicting numerical values and binary classifications, such as determining whether a patient has low or high heart attack risk.

Clustering Techniques (K-Means, Hierarchical Clustering): Used in grouping patients with similar health conditions to predict risk categories effectively.

Bayesian Models (Naïve Bayes, Bayesian Networks): Used in probabilistic risk assessments, helping in early disease detection and personalized treatment recommendations.

K-Nearest Neighbors (KNN): Identifies similar patient cases based on existing health records, providing insights into potential risks.

Support Vector Machines (SVMs): Create hyperplanes to classify patients based on heart disease symptoms and risk factors.

Ensemble Learning (Random Forest, Bagging, Boosting): Combines multiple models to enhance prediction accuracy. Random Forest, a widely used technique, builds multiple decision trees and combines results to improve risk assessment reliability.

3. Need for AI and Machine Learning

The rapid increase in heart disease cases worldwide has highlighted the urgent need for predictive healthcare solutions. Traditional methods rely on medical expertise and historical patient data, but ML-based predictive models offer faster, more accurate, and scalable solutions. AI-powered systems can:

- Analyze large volumes of patient data and detect hidden patterns.
- Identify high-risk individuals based on multiple health parameters.
- Provide early warning systems for healthcare providers.
- Improve clinical decision-making, reducing dependency on manual diagnostics.
- Optimize treatment plans and preventive care strategies.

1.2 Problem on Hand

Despite advancements in machine learning (ML) for heart disease prediction, existing studies face challenges in accuracy, optimization, and real-world validation. Dubey et al. (2023) found SVM (93%) and Logistic Regression (89%) to be effective, while Sahoo et al. (2023) reported Random Forest (90.16%) as the best model. Marqas et al. (2023) achieved 91.91% accuracy using XGBoost, but lacked real-world testing. Drod et al. (2022) used Logistic Regression and PCA (AUC 0.87) but focused only on MAFLD patients.

Most studies focus on specific ML techniques without optimizing feature selection or validating results with real-world data. This study addresses these gaps by developing a Logistic Regression-based predictive model while integrating survey data from 500 respondents to enhance accuracy and clinical applicability.

1.3 Importance of the Problem

The existing challenges in heart disease prediction highlight the need for more accurate and efficient machine learning models**.** Addressing these issues can improve early detection, patient outcomes, and healthcare efficiency**.**

- Existing heart disease prediction models lack accuracy, leading to high false positives and false negatives and increasing the risk of misdiagnosis.
- Most studies focus only on individual ML techniques without integrating optimization algorithms to improve predictive performance.
- There is a lack of research on ensemble learning methods such as Random Forest, XGBoost, and AdaBoost, which can enhance model stability and accuracy.
- Traditional diagnostic methods are time-consuming and expensive, making early detection difficult, whereas ML-based predictive models can help identify high-risk individuals sooner.
- Implementing an optimized ML model can assist doctors in clinical decision-making, provide real-time heart health monitoring, and reduce healthcare costs.

1.4 Scope of the Study

This research aims to develop an optimized heart disease prediction model using Logistic Regression to improve diagnostic accuracy and early risk assessment. By leveraging machine learning in predictive healthcare, the study enables data-driven decision-making for timely intervention.

The proposed model offers a computationally efficient and interpretable approach, ensuring easy integration into real-world medical practices. This research will contribute to early diagnosis, reduced mortality rates, and improved patient outcomes, advancing AI-driven predictive analytics in healthcare.

CHAPTER- 2

LITERATURE REVIEW

Raviprakash et al. (2024) explored various machine learning (ML) models for heart disease prediction, focusing on ensemble learning and deep learning methods. The study compared traditional ML models like Support Vector Machines (SVM) and Random Forest, which have been widely used for cardiac abnormality detection, with advanced deep learning models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN). The results showed that CNN and RNN achieved superior accuracy by capturing intricate patterns and dependencies in medical data. However, the study highlighted overfitting challenges in deep learning models and emphasized the need for feature selection and optimization techniques to improve predictive performance.

Marqas et al. (2023) developed a heart attack risk prediction model using real-world patient data, incorporating demographics, medical history, and lifestyle factors. The study applied Logistic Regression, Decision Trees, and Random Forest, evaluating them using accuracy, precision, recall, and F1-score. Among the models, XGBoost achieved the highest accuracy of 91.91%, while the Extra Tree Classifier recorded the highest AUC of 0.950. The study proposed the integration of ML models with Electronic Health Records (EHRs) for real-time clinical decision-making. However, real-world testing and model explainability were identified as gaps that need further exploration.

García-Ordás et al. (2023) investigated deep learning-based heart disease risk prediction, using Sparse Autoencoders (SAE) and Convolutional Neural Networks (CNNs) to improve feature extraction and classification. The feature augmentation approach significantly enhanced model performance, with CNNs achieving a precision rate of 90%. Compared to traditional ML techniques, the deep learning model was better at capturing complex risk patterns. Despite the promising results, the study acknowledged high computational costs and the need for larger datasets for effective implementation.

Dubey A. K. et al. (2023) conducted a comparative analysis of ML models, including Logistic Regression, Decision Trees, Random Forest, and SVM, using Cleveland and Statlog datasets. The study found that SVM performed best with an accuracy of 93%, while Logistic Regression achieved 89% accuracy. The authors emphasized that feature selection and preprocessing techniques play a crucial role in improving model reliability. However, they noted a lack of real-world validation and interpretability in existing ML-based heart disease prediction models.

Sahoo G. K. et al. (2023) compared various ML models, including Logistic Regression, K-Nearest Neighbors (KNN), SVM, Naïve Bayes, Decision Trees, Random Forest, and XGBoost, using the UCI heart disease dataset. Their results showed that Random Forest provided the highest accuracy at 90.16%, making it a reliable model for heart disease classification. However, the study lacked a focus on feature optimization techniques that could further enhance prediction accuracy.

Drod et al. (2022) applied ML techniques to identify cardiovascular disease (CVD) risk factors in 191 metabolic-associated fatty liver disease (MAFLD) patients. The study used Logistic Regression, Univariate Feature Ranking, and Principal Component Analysis (PCA) to determine key risk factors, identifying hypercholesterolemia, plaque scores, and diabetes duration as the most significant contributors to heart disease. The ML model achieved 85.11% accuracy for high-risk cases and 79.17% for low-risk cases, with an AUC of 0.87. However, the study was limited to MAFLD patients, reducing its generalizability.

Gupta et al. (2021) developed an ML-based heart attack prediction model using Gradient Boosting, Decision Trees, Random Forest, and Logistic Regression, utilizing datasets from the Framingham study and the UCI Heart repository. The study found that Gradient Boosting achieved the highest accuracy, with key influencing factors being chest pain type, cholesterol levels, heart rate, thalassemia, and age. The research emphasized the importance of lifestyle factors in preventing premature heart attacks and suggested further studies on hybrid ML models.

Suraj Kumar Gupta et al. (2021) proposed a Machine Learning-based Heart Attack Prediction model using Gradient Boosting, demonstrating high accuracy in predicting heart attacks. The authors emphasized that AI-driven models could prevent up to 80% of premature heart attack cases by enabling early detection and lifestyle modifications. However, the study lacked insights into model interpretability and clinical validation.

Khourdifi and Bahaj (2019) developed a hybrid ML model for heart disease prediction using Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) for feature selection. The study utilized Fast Correlation-Based Feature Selection (FCBF) and classified patients using KNN, SVM, Naïve Bayes, Random Forest, and Artificial Neural Networks (ANNs). The hybrid optimization approach achieved an accuracy of 99.65%, highlighting its effectiveness. However, the study lacked real-world clinical validation and practical applicability in healthcare settings.

Ahmadian et al. (2019) proposed a hybrid ML model integrating clinical parameters and ECG measurements for cardiovascular disease prediction. The study combined Random Forest and deep learning techniques, achieving 89% accuracy. While the model showed strong predictive capabilities, it lacked integration with real-world patient monitoring systems, limiting its broader usability.

2.1 Research gap identification

Author(s) & Year	Objective of the Study	Identified Research Gap
Raviprakash et al. (2024)	Explored ML models for heart disease prediction using ensemble learning and deep learning techniques.	Lack of focus on real-time implementation of ensemble learning models for clinical applications.
Marqas et al. (2023)	Developed a heart attack risk prediction model using real-world patient data and evaluated various ML models.	Limited interpretability and explainability of ML models in healthcare decision-making.
García-Ordás et al. (2023)	Investigated deep learning-based heart disease risk prediction using feature augmentation techniques.	Lack of comparative studies between deep learning and hybrid ML models for heart disease prediction.
Dubey A. K. et al. (2023)	Conducted a comparative analysis of ML models for heart disease prediction.	Lack of integration of real-time patient monitoring for continuous risk assessment.
Sahoo G. K. et al. (2023)	Compared various ML models using the UCI heart disease dataset.	Limited focus on feature engineering and optimization for improved predictive accuracy.
Drod et al. (2022)	Identified cardiovascular disease (CVD) risk factors in MAFLD patients using ML techniques.	Lack of generalization across diverse patient populations, as the study was limited to MAFLD patients.
Gupta et al. (2021)	Developed an ML-based heart attack prediction model using supervised learning classifiers.	Limited studies on ensemble-based optimization techniques for improving prediction accuracy.
Suraj Kumar Gupta et al. (2021)	Developed a Gradient Boosting-based ML model for heart attack prediction.	Lack of explainability and transparency in AI-driven predictions for clinical use.
Khourdifi and Bahaj (2019)	Developed a hybrid ML model using PSO and ACO for feature selection in heart disease classification.	Need for real-world validation of the proposed optimization techniques in clinical datasets.
Ahmadian et al. (2019)	Employed a hybrid ML model integrating clinical parameters and ECG for cardiovascular disease prediction.	Limited integration of deep learning techniques with ML models for improved classification performance.

2.2 Flow of the Study

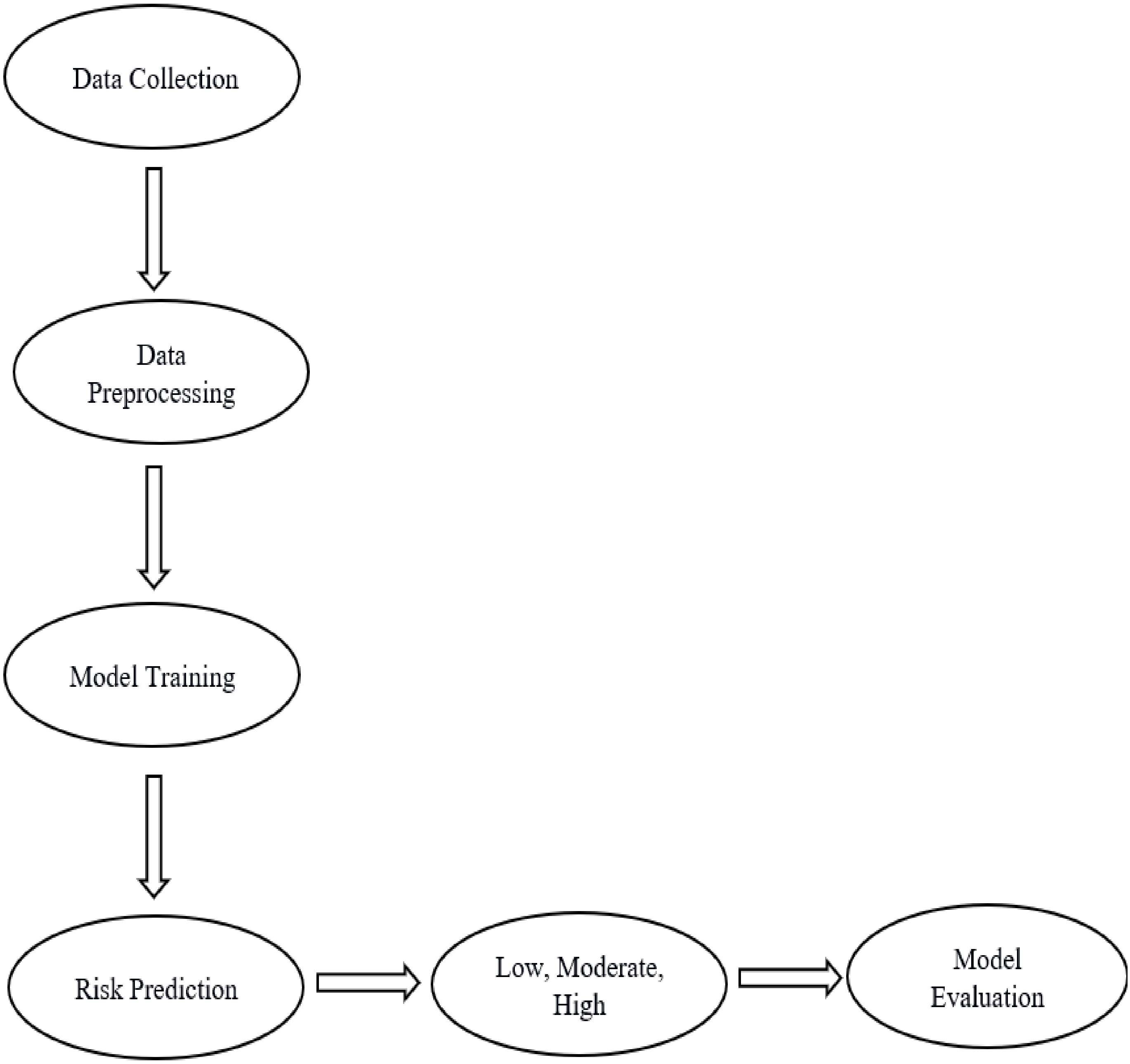

Fig.1.1 Flow of the Study

2.3 Research questions

- What are heart attack risk factors, and how do they affect how well machine learning models forecast results?

- How may logistic regression and ensemble learning improve the precision and dependability of heart attack risk assessment?

- What obstacles need to be removed before clinical practice can effectively include machine learning-based models for predicting heart disease?

2.4 Objective of the study

- To optimize a risk prediction model using logistic regression, improving classification accuracy and reliability for heart attack detection.

- To improve predictive accuracy and fill optimization gaps to improve feature selection and model performance.

- To investigate if incorporating machine learning (ML)-based heart attack prediction models into practical clinical applications for early detection and prevention is feasible.

CHAPTER- 3

RESEARCH METHODOLOGY

3.1 Research Design

This research uses primary and secondary data sources to create a thorough model for predicting the risk of a heart attack. A strong, data-driven method for examining heart disease risk factors is ensured by the study's integration of structured machine learning datasets (secondary data) with real-world survey responses (primary data).

Primary Data

Primary data refers to first-hand information collected directly from respondents for a specific research purpose. In this study, a structured survey of 500 respondents serves as the primary data source. The survey includes closed-ended and Likert-scale questions, covering demographics, lifestyle habits, medical history, stress levels, and healthcare awareness. The objective is to capture real-world behavioral and environmental risk factors contributing to heart disease. The collected responses help in validating the ML model's predictions, ensuring that the study considers both clinical indicators and human-centered insights.

Secondary Data

Secondary data refers to pre-existing datasets collected from external sources. This study utilizes a heart attack risk prediction dataset from Kaggle, which includes 50,000 patient records with clinical and demographic features such as age, cholesterol levels, blood pressure, diabetes history, smoking habits, BMI, and ECG results. This structured dataset enables the training and testing of a Logistic Regression model, improving the accuracy and automation of heart attack risk classification. Additionally, literature reviews from Google Scholar and ScienceDirect provide insights into previous ML models, risk factors, and advancements in predictive healthcare.

Patient ID	Age	Sex	Cholesterol	Blood Pressure	Heart Rate	Diabetes	Family History	Smoking	Obesity	Alcohol Consumption	Exercise Hours Per Week	Diet	Previous Heart Problems	Medication Use	Stress L
BMW7812	67	Male	208	158/88	72	0	0	1	0	0	4.168188835	Average	0	0	
CZE1114	21	Male	389	165/93	98	1	1	1	1	1	1.813241618	Unhealthy	1	0	
BNI9906	21	Female	324	174/99	72	1	0	0	0	0	2.078352986	Healthy	1	1	
JLN3497	84	Male	383	163/100	73	1	1	1	0	1	9.828129593	Average	1	0	
GFO8847	66	Male	318	91/88	93	1	1	1	1	0	5.80429882	Unhealthy	1	0	
ZOO7941	54	Female	297	172/86	48	1	1	1	0	1	0.625008024	Unhealthy	1	1	
WYV0966	90	Male	358	102/73	84	0	0	1	0	1	4.098177091	Healthy	0	0	
XXM0972	84	Male	220	131/68	107	0	0	1	1	1	3.427928754	Average	0	1	
XCQ5937	20	Male	145	144/105	68	1	0	1	1	0	16.86830224	Average	0	0	
FTJ5456	43	Female	248	160/70	55	0	1	1	1	1	0.194515061	Unhealthy	0	0	
HSD6283	73	Female	373	107/69	97	1	1	1	0	1	16.84198759	Average	1	1	
YSP0073	71	Male	374	158/71	70	1	1	1	1	1	8.251995072	Average	0	0	
FPS0415	77	Male	228	101/72	68	1	1	1	1	1	19.63326816	Unhealthy	0	0	
YYU9565	60	Male	259	169/72	85	1	1	1	0	1	17.03737418	Healthy	1	1	
VTW9069	88	Male	297	112/81	102	1	1	1	0	1	15.38760463	Unhealthy	0	1	
DCY3282	73	Male	122	114/88	97	1	1	1	0	1	14.5596645	Average	0	0	
DXB2434	69	Male	379	173/75	40	1	1	1	1	1	4.184648046	Average	1	0	
COP0566	38	Male	166	120/74	56	1	0	1	1	0	8.91790723	Healthy	0	1	
XBI0592	50	Female	303	120/100	104	1	0	1	0	1	4.943580247	Average	1	1	
RQX1211	60	Male	145	160/98	71	1	0	1	0	1	1.89255933	Healthy	1	0	
MBI0008	66	Male	340	180/101	69	1	0	1	1	0	9.105433807	Unhealthy	1	1	
RVN4963	45	Male	294	130/84	66	0	0	1	1	1	13.69378493	Healthy	0	0	
LBY7992	50	Male	359	175/60	97	0	1	1	0	1	8.354399413	Healthy	1	0	
RDI3071	84	Male	202	173/109	81	1	1	1	0	1	10.99581328	Unhealthy	1	0	
NCU1956	36	Male	133	161/90	97	1	0	1	1	1	3.61800905	Healthy	1	0	
MSW4208	90	Male	159	140/95	52	0	0	1	0	1	10.7098555	Healthy	0	1	

The first phase focuses on building a predictive model using Logistic Regression to classify individuals into Low, Moderate, or High heart attack risk categories. This phase consists of the following key steps:

Data Import and Preprocessing: The dataset, obtained from Kaggle, is imported and examined for missing values, inconsistencies, and duplicate records. Missing values are handled using mean/mode imputation or median replacement, while categorical variables are encoded using label encoding or one-hot encoding. Numerical features such as cholesterol levels, BMI, and blood pressure are normalized to improve model performance.

Feature Selection: To enhance model accuracy and reduce noise, correlation analysis and Recursive Feature Elimination (RFE) are applied. This step ensures that only the most relevant variables contribute to heart attack prediction.

Data Splitting and Model Training: The dataset is split into 80% training data and 20% testing data, ensuring a balanced distribution of target classes. The Logistic Regression model is trained using hyperparameter tuning techniques, such as grid search and cross-validation, to improve prediction accuracy.

Model Evaluation: The trained model is tested on the 20% testing dataset, and its performance is assessed using accuracy, precision, recall, F1-score, and AUC-ROC curve analysis. These metrics help evaluate the model's ability to correctly classify heart attack risk levels.

Model Optimization: If the model underperforms, adjustments such as feature engineering, data resampling and parameter fine-tuning are performed to enhance accuracy and reliability.

Phase 2: Survey-Based Analysis

To complement the ML model and validate its findings, a survey of 500 respondents is conducted, focusing on real-world risk factors that influence heart attack probability.

Survey Design & Data Collection: A structured questionnaire is developed covering demographics, medical history, lifestyle habits, behavioral patterns, and healthcare awareness. The survey is conducted through online and offline channels to ensure a diverse and representative sample.

Statistical Analysis of Survey Data: The collected responses are analyzed using descriptive statistics, correlation tests, and regression analysis to identify significant predictors of heart disease.

Comparison with ML Model Predictions: The survey findings are cross-validated against the ML model's predictions, allowing us to assess how well machine learning aligns with real-world observations.

3.2 Data Source

To develop an accurate and reliable heart attack risk prediction model, this study utilizes two primary data sources:

Machine Learning Dataset (Kaggle): A structured dataset from Kaggle containing 50,000 records with clinical and demographic attributes such as age, cholesterol level, blood pressure, diabetes history, smoking habits, BMI, stress levels, ECG results, and heart rate. This dataset is used for training and validating the Logistic Regression model, enabling automated risk classification.

Literature Review Sources: To support the study with scientific evidence and best practices, relevant research papers from Google Scholar, ScienceDirect, PubMed, and IEEE Xplore are reviewed. These sources provide valuable insights into previous ML models, heart disease risk factors, and advancements in AI-driven healthcare.

3.3 Sample size

To ensure a robust and reliable heart attack risk prediction model, this study leverages a large dataset alongside real-world survey responses. This dual approach integrates machine learning (ML) techniques with human-centered insights, providing a well-rounded, data-driven framework for heart disease risk assessment.

The Heart Attack Risk Prediction dataset, obtained from Kaggle, comprises 50,000 patient records, making it a large-scale, diverse dataset ideal for training and testing the Logistic Regression model. The dataset encompasses 20 key features, which are categorized as follows:

The dataset used in this study consists of various features categorized into **demographics, lifestyle factors, medical indicators, and cardiac health parameters**, all of which play a crucial role in predicting heart attack risk.

Demographic factors include **age, gender, and family history**. As age increases, the risk of heart disease rises, making it an important predictor. Gender also plays a significant role, as men generally have a higher risk at younger ages compared to women. Additionally, a family history of heart disease indicates a genetic predisposition, which increases the likelihood of developing cardiovascular conditions.

Lifestyle factors such as **smoking, alcohol consumption, and physical activity level** contribute significantly to heart attack risk. Tobacco consumption damages arteries and raises the risk of atherosclerosis, while excessive alcohol intake can lead to high blood pressure and irregular heart rhythms. On the other hand, maintaining an active lifestyle helps prevent obesity, diabetes, and hypertension, all of which are major contributors to heart disease.

Medical indicators such as **diabetes, hypertension, cholesterol levels, and blood pressure** are essential in evaluating cardiovascular health. High blood sugar levels can damage arteries over time, increasing the risk of heart attacks. Hypertension forces the heart to work harder than normal, leading to long-term complications. Elevated cholesterol levels contribute to plaque buildup in arteries, restricting blood flow, while high resting blood pressure is a sign of cardiovascular stress. Additionally, fasting blood sugar levels are a critical indicator of diabetes, a condition closely linked to heart disease.

Cardiac health parameters provide deeper insights into cardiovascular function. **BMI (Body Mass Index)** is a key measure of obesity, a well-established risk factor for heart disease. **Heart rate** and **stress levels** influence overall heart function, as chronic stress contributes to high blood pressure and inflammation. **Chest pain type** helps classify different heart conditions, while **thalassemia**, a genetic blood disorder, can also contribute to heart complications. **ECG results** provide crucial information on heart rhythm abnormalities, and **exercise-induced angina** signals potential coronary artery disease. Additionally, the **maximum heart rate achieved** helps assess cardiovascular fitness.

The **target variable in this study is heart attack risk**, which is classified into **low, moderate, or high risk**. This classification allows the machine learning model to effectively predict the likelihood of a heart attack based on the combined influence of clinical, demographic, and lifestyle-related factors. Through the integration of these features, this study aims to develop a data-driven, interpretable, and accurate heart attack risk prediction model.

The large dataset size ensures that the model is trained on diverse patient data, improving its generalizability, accuracy, and reliability in predicting heart attack risks. It also provides a balanced representation of different risk factors, demographic groups, and medical conditions, reducing bias and improving real-world applicability.

To complement the ML-based predictions, a structured survey was conducted with 500 respondents to analyze real-world risk factors and validate the model's findings. The survey was carefully designed to capture both medical and behavioral aspects of heart disease risk, ensuring a comprehensive assessment of factors influencing cardiovascular health.

The demographic section of the survey focused on **age, gender, and family history of heart disease**. Understanding the distribution of heart disease risk across different **age groups and genders** helps identify population segments that are more vulnerable to cardiovascular issues. A family history of heart disease is a critical factor, as genetic predisposition significantly increases an individual's risk of developing cardiac complications.

Lifestyle factors were assessed through questions related to **smoking, alcohol consumption, diet, physical activity, and sleep patterns**. The survey examined the **impact of tobacco and alcohol use** on cardiovascular health, as both substances are known contributors to heart disease. Participants were also asked about their **diet and nutrition**, particularly their intake of high-fat, high-sugar, and processed foods, which are major risk factors for obesity and cholesterol-related heart conditions. Additionally, the survey captured data on **exercise habits and physical activity levels**, evaluating how frequently individuals engage in cardio-protective activities such as running, walking, or gym workouts. Since **poor sleep quality and irregular sleep cycles** can lead to high blood pressure and other health issues, sleep patterns were also analyzed.

The medical history section aimed to assess the **prevalence of hypertension, diabetes, cholesterol levels, and prior heart conditions** among respondents. High blood pressure and diabetes are **leading causes of cardiovascular disease**, making it essential to investigate their occurrence in the surveyed population. Similarly, **cholesterol levels were examined**, as they are heavily influenced by dietary habits and genetics. Participants were also asked whether they had previously experienced **heart attacks, angina, or other cardiac complications**, helping to establish connections between existing medical conditions and potential future risks.

Behavioral factors such as **stress levels, work-life balance, and mental health** were also included in the survey. The role of work-related stress, financial pressures, and emotional well-being in cardiovascular health was examined, as chronic stress can contribute to **hypertension and heart disease**. Questions regarding **work-life balance** explored how long working hours, lack of relaxation, and excessive workload may impact heart health. Additionally, mental health conditions, such as anxiety and depression, were analyzed to understand their correlation with heart disease.

Lastly, the survey assessed **healthcare awareness and accessibility**, focusing on **regular medical check-ups, knowledge of heart disease prevention, and access to healthcare facilities**. The frequency of **preventive screenings and medical consultations** was evaluated to determine how proactive respondents were about their heart health. Their **awareness of preventive measures**, including lifestyle modifications and early symptom recognition, was also measured. Furthermore, questions about **access to healthcare facilities** explored whether **income levels, geographical location, or availability of medical services** impacted individuals' ability to manage their heart health effectively.

By integrating **survey responses with ML predictions**, this study ensures that **heart attack risk assessment** is not only based on clinical data but also considers **behavioral, lifestyle, and healthcare accessibility factors**, making the findings more **practical and applicable in real-world scenarios**.

CHAPTER – 4

DATA COLLECTION, ANALYSIS & INTERPRETATION

4.1 Type of Data Required

Predicting heart attack risk requires a dataset that captures the multidimensional nature of cardiovascular health, since heart disease is not caused by a single factor but by a combination of physiological, behavioural, demographic, and hereditary influences. Therefore, the type of data required for this study must be comprehensive enough to represent real-world patient health conditions and suitable for analytical and predictive modeling using logistic regression.

➢ **Demographic Data**

Demographic variables form the foundation for understanding risk variations across age groups and gender.
The necessary demographic attributes include:

- **Age:** A primary determinant of cardiovascular vulnerability.
- **Sex:** Differences in male/female risk patterns often influence model interpretation.

Demographic data helps in segmenting patients and identifying population-specific risk trends.

➢ **Clinical and Physiological Data**

Cardiovascular diseases are strongly associated with measurable physiological indicators. Therefore, the dataset requires detailed clinical parameters such as:

- Cholesterol levels (total cholesterol)
- Triglycerides
- BMI (Body Mass Index)
- Blood Pressure (Systolic & Diastolic)
- Resting Heart Rate

These variables reflect metabolic health, arterial condition, plaque formation tendencies, and cardiac function. They are essential for developing a medically meaningful and statistically significant model.

➢ **Medical History and Pre-existing Conditions**

A patient’s medical background significantly affects future cardiac risk. Thus, data is required on:

- Diabetes status
- Previous heart problems
- Medication usage
- Family history of heart disease

These factors help the model understand chronic conditions and inherited risks, which strongly influence cardiovascular outcomes.

➢ **Lifestyle and Behavioural Data**

Lifestyle choices directly impact heart health and are often responsible for long-term risk accumulation. The study requires data related to:

- Smoking habits
- Alcohol consumption
- Exercise hours per week
- Sedentary hours per day
- Diet quality (healthy/average/unhealthy)
- Sleep duration
- Stress level

These variables help quantify the behavioural influences on heart disease and contribute significantly to predictive accuracy.

➢ **Socio-economic Indicators**

While not the strongest medical predictors, factors such as income level contribute to differences in healthcare access, lifestyle patterns, dietary quality, and stress. Including such variables improves holistic model understanding.

➢ **Output Variable (Dependent Variable)**

The essential requirement for logistic regression is a binary outcome variable, hence:

- **Heart Attack Risk:** Classified as **High** or **Low** based on clinical thresholds.

This outcome variable enables the model to assign a probability of risk to each individual.

4.2 Sources of Data

The present study is based entirely on secondary data, which was obtained from publicly accessible digital repositories that compile medical and health-related datasets. Since the study involves the prediction of heart attack risk, the use of a large, structured, and reliable dataset was essential to ensure the accuracy and validity of the logistic regression model. Secondary data was preferred because it provides extensive patient records, a wide range of clinical variables, and consistency in data collection—all of which are crucial for machine-learning analysis.

1. **Source: Online Public Health Dataset (Kaggle Repository)**

    The main dataset used for this study was sourced from Kaggle, one of the largest open-source platforms for data science and machine learning. The dataset titled *"Heart Attack Risk Prediction Dataset"* contains 50,000 anonymized patient records with 26 variables related to demographic details, lifestyle patterns, clinical measurements, and medical history.

    This dataset was selected due to the following reasons:

- It offers a sufficiently large sample size, necessary for predictive modeling.
- It includes multiple risk factors known to affect cardiovascular health.
- The data is already structured and standardized, making it suitable for logistic regression.
- Anonymization ensures compliance with ethical norms and protects patient confidentiality.
- Widely used in academic and research projects, indicating reliability.

In addition to the main dataset, information from various academic sources was consulted to better understand heart attack risk factors and justify the selection of variables used for the model. These include:

- Peer-reviewed journals
- Clinical research studies
- WHO and CDC cardiovascular health publications
- Medical literature from Google Scholar, ResearchGate, and PubMed

These sources were not used as datasets, but they helped validate the medical significance of variables such as cholesterol, BP, BMI, diabetes, and lifestyle habits.

2. Rationale for Using Secondary Data

The use of secondary data was appropriate for the study because:

- Collecting primary medical data requires ethical permission, clinical involvement, and patient participation.
- The scale of primary data collection (50,000+ records) is impractical in an academic timeframe.
- Secondary datasets provide diverse and standardized measurements, essential for machine learning.
- The dataset used already contains a broad spectrum of health indicators, eliminating the need for manual data recording.

3. Reliability and Authenticity of Data Source

Kaggle datasets are contributed by domain professionals and are widely used for:

- Academic research
- Predictive modeling competitions
- Healthcare analytics projects

The dataset used in this study has been previously cited by other research practitioners, increasing confidence in its quality and generalizability.

4.3 Descriptive Statistics

Descriptive statistics were computed to understand the structure, distribution, and characteristics of the dataset before conducting further analytical procedures. Since heart attack prediction involves diverse clinical, demographic, and lifestyle-related parameters, descriptive statistics helped identify trends, patterns, outliers, and preliminary relationships among variables.

The dataset consisted of 50,000 patient records and 26 variables, which included demographic information (Age, Sex), lifestyle behaviours (Smoking, Alcohol Use, Exercise Hours, Diet), clinical metrics (BMI, Cholesterol, Triglycerides, Blood Pressure, Heart Rate), medical history (Diabetes, Previous Heart Conditions, Medication Use), and the outcome variable Heart Attack Risk.

1. Distribution of the Dependent Variable (Heart Attack Risk)

The target variable, *Heart Attack Risk*, was originally categorical (Low/High), later encoded into binary form:

- **0 → Low Risk**
- **1 → High Risk**

Frequency Distribution

Risk Category	Count
Low Risk (0)	25,761
High Risk (1)	24,239

The distribution reveals a **balanced dataset**, ensuring that the logistic regression model does not suffer from class imbalance. This balance improves model accuracy and stability.

2. Descriptive Summary of Demographic Variables

Age

- **Range:** 18 to 95 years
- **Mean:** Around middle age
- **Observation:** Higher risk individuals tended to fall within older age groups, consistent with medical literature.

Sex

- Distributed across male and female categories.
- Used for encoding to maintain model interpretability.

Demographic attributes help in identifying population-level patterns and risk burdens, which are crucial for segmentation and prediction.

3. Clinical Variables

These include measurable physiological indicators directly associated with cardiovascular health.

Cholesterol

- Significant variation observed across patients.

- Many individuals displayed elevated levels, indicating metabolic imbalance.
- Cholesterol showed a strong positive correlation with risk.

Triglycerides

- Higher triglycerides were commonly found among high-risk patients.
- Strong clinical relevance due to their role in plaque formation.

BMI (Body Mass Index)

- Majority of the sample fell into overweight and obese categories.
- Obesity is known to amplify risk for hypertension, diabetes, and cardiac issues.

Blood Pressure (Systolic & Diastolic)

- Originally recorded as “120/80”, split into separate variables.
- **Systolic BP** showed greater variation and stronger correlation with risk.
- High systolic pressure is a medical indicator of cardiovascular strain.

Heart Rate

- Resting heart rate values varied significantly.
- Higher heart rates observed more frequently among high-risk individuals.

These clinical indicators provide the strongest predictive power in medical risk assessment.

4. Lifestyle and Behavioural Variables

Lifestyle attributes form an essential component of cardiac risk prediction since a significant portion of heart disease is lifestyle-induced.

Smoking

- Clear risk escalation observed among smokers.
- A strong behavioural predictor aligned with global research.

Alcohol Consumption

- Moderate-to-high intake linked with metabolic issues and elevated risk.

Exercise Hours per Week

- Lower physical activity observed among high-risk individuals.

- Demonstrates protective effect of routine exercise.

Sedentary Hours

- More sedentary hours correlated with higher BMI and increased risk.

Diet

- Encoded as Healthy, Average, Unhealthy.
- Poor diet strongly associated with high-risk category.

Sleep Hours

- Insufficient sleep is associated with stress and elevated BP.

Behavioural variables enrich the dataset with holistic health information that complements clinical metrics.

5. Medical History Variables

These include chronic conditions and hereditary factors influencing risk.

Diabetes

- One of the strongest indicators of high-risk status.
- A significantly higher proportion of diabetics belonged to the high-risk category.

Previous Heart Problems

- Strongly correlated with high-risk predictions, as expected.

Medication Use

- Indicates ongoing medical treatment or chronic health issues.

Family History

- Hereditary predisposition increased probability of high risk.

These variables strengthen the predictive model by accounting for long-term health factors.

6. Visual Descriptive Analysis

Several visual tools were used to summarize and explore the dataset:

Boxplot (Age vs Heart Attack Risk)

- Showed that median age is higher among high-risk individuals.
- Displayed wider spread for high-risk group.

Bar Graph (Diabetes vs Heart Attack Risk)

- Revealed a clear association between diabetes and high risk.
- Validated diabetes as a crucial predictor.

Correlation Heatmap

- Showed strong positive correlations between:
  - ✓ Cholesterol & Triglycerides
  - ✓ Systolic BP & Diastolic BP
  - ✓ BMI & Triglycerides
- Confirmed absence of harmful multicollinearity among major variables, supporting logistic regression suitabilility.

4.4 Processing of data for analysis

Data processing is a critical step in preparing the dataset for meaningful statistical analysis and predictive modeling. Since the study is based on a secondary dataset of 50,000 patient records, several preprocessing procedures were applied to ensure accuracy, consistency, and suitability for logistic regression. The following steps outline the entire data-processing workflow undertaken before model development.

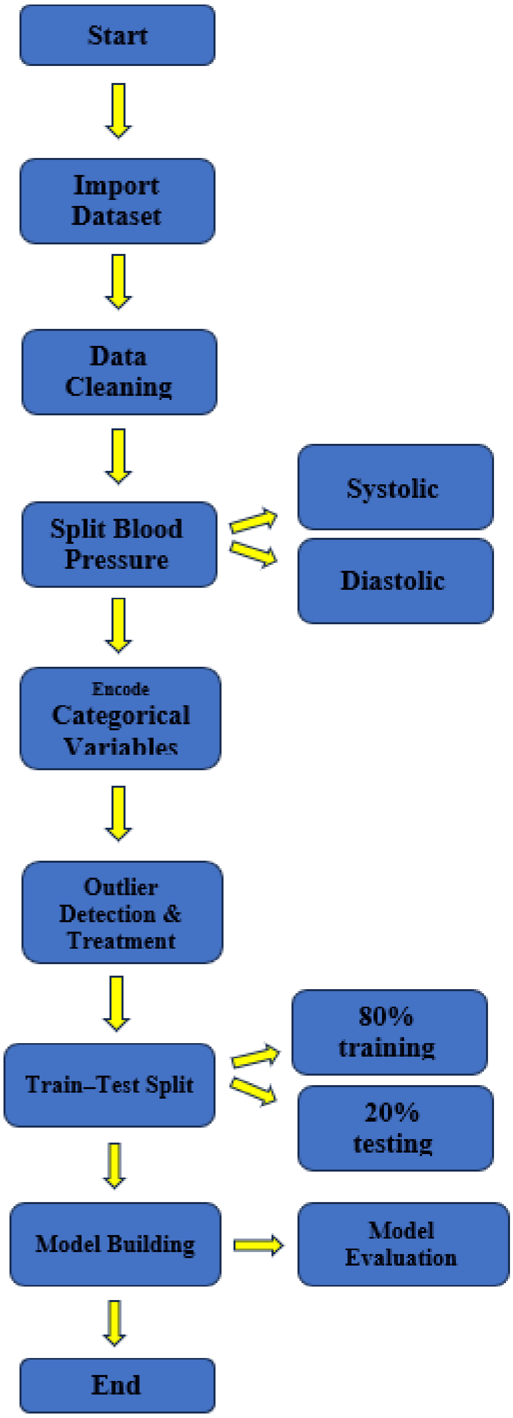

Fig.1.2 Heart Attack Risk Prediction Flowchart

4.5 Analysis of data

This section presents a systematic analysis of the processed dataset to understand its structure, key patterns, and the influence of various health indicators on heart attack risk. Since the study relies exclusively on secondary data comprising 50,000 patient health records, exploratory data analysis (EDA) and logistic regression were used to derive meaningful insights and develop a predictive model. All statistical computations and visualisations were performed using Python.Several statistical and visualization methods were applied to uncover the structure and distribution of the data:

4.5.1 Exploratory Data Analysis (EDA)

The EDA stage provided an initial understanding of how demographic, medical, and lifestyle variables behave within the dataset. A combination of univariate, bivariate, and correlation-based analyses was conducted.

a. Univariate Analysis

Each variable was examined individually to understand its distribution and spread.

- Age ranged from young adults to the elderly, indicating a diverse sample.
- Cholesterol, Triglycerides, and BMI displayed positively skewed distributions, which is common in medical data where extreme values occur due to health conditions.
- Heart Attack Risk was nearly balanced, with 25,761 low-risk and 24,239 high-risk individuals.
- Diabetes, Smoking, and Family History showed notable presence across the dataset, suggesting their importance in cardiovascular profiling.

These findings highlighted the relevance of both lifestyle and clinical factors in determining cardiac risk.

b. Bivariate Analysis

Bivariate analysis explored the relationship between predictor variables and the target outcome. Various visualisations were used to illustrate these patterns.

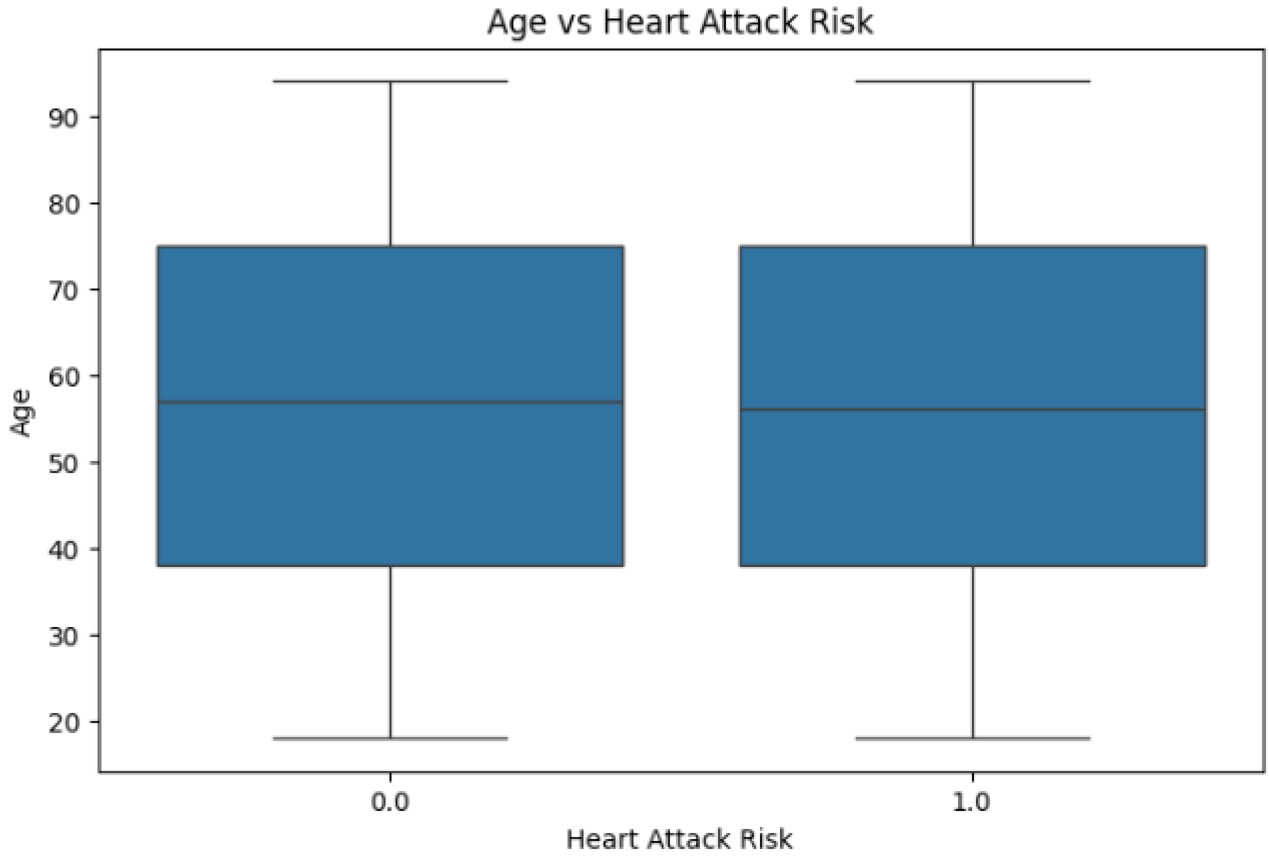

Fig.1.3 Age vs Heart Attack Risk

The boxplot comparing age across low-risk (0) and high-risk (1) categories reveals that the distribution of age is quite similar for both groups. The median age for low-risk and high-risk individuals is almost the same, and the interquartile ranges show substantial overlap. This suggests that age, although clinically relevant, does not independently differentiate risk levels in this dataset.

The presence of both young and older individuals in both categories indicates that heart attack risk is influenced by other factors such as diabetes, blood pressure, cholesterol, lifestyle patterns, and BMI. Therefore, while age contributes to risk, it does not act as a strong standalone predictor. This supports the need for a multivariate prediction model like logistic regression, which can combine age with other risk factors to improve classification accuracy.

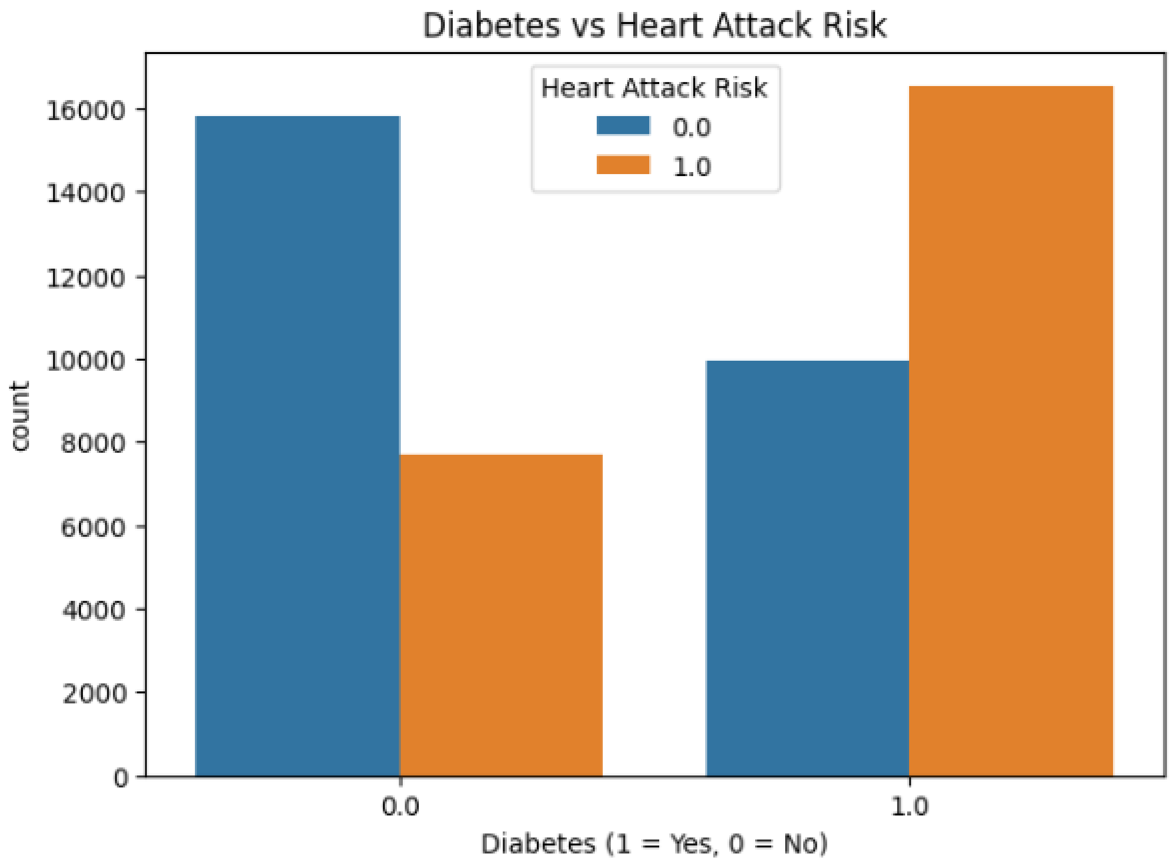

Fig.1.4 Diabetes vs Heart Attack Risk

The bar chart clearly shows a strong relationship between diabetes and heart attack risk. Among individuals **without diabetes (0)**, the majority fall in the low-risk category. In contrast, among those **with diabetes (1)**, the high-risk category dominates significantly.

This indicates that **diabetes is one of the strongest determinants of heart attack risk** in the dataset. Clinically, this aligns with existing literature stating that diabetes accelerates cardiovascular damage, increases cholesterol abnormalities, and contributes to hypertension, all of which elevate heart attack risk.

Therefore, diabetes emerges as a **key predictor** in the logistic regression model. Its clear separation pattern also explains why models including diabetes as a feature often achieve higher accuracy.

c. Correlation Analysis

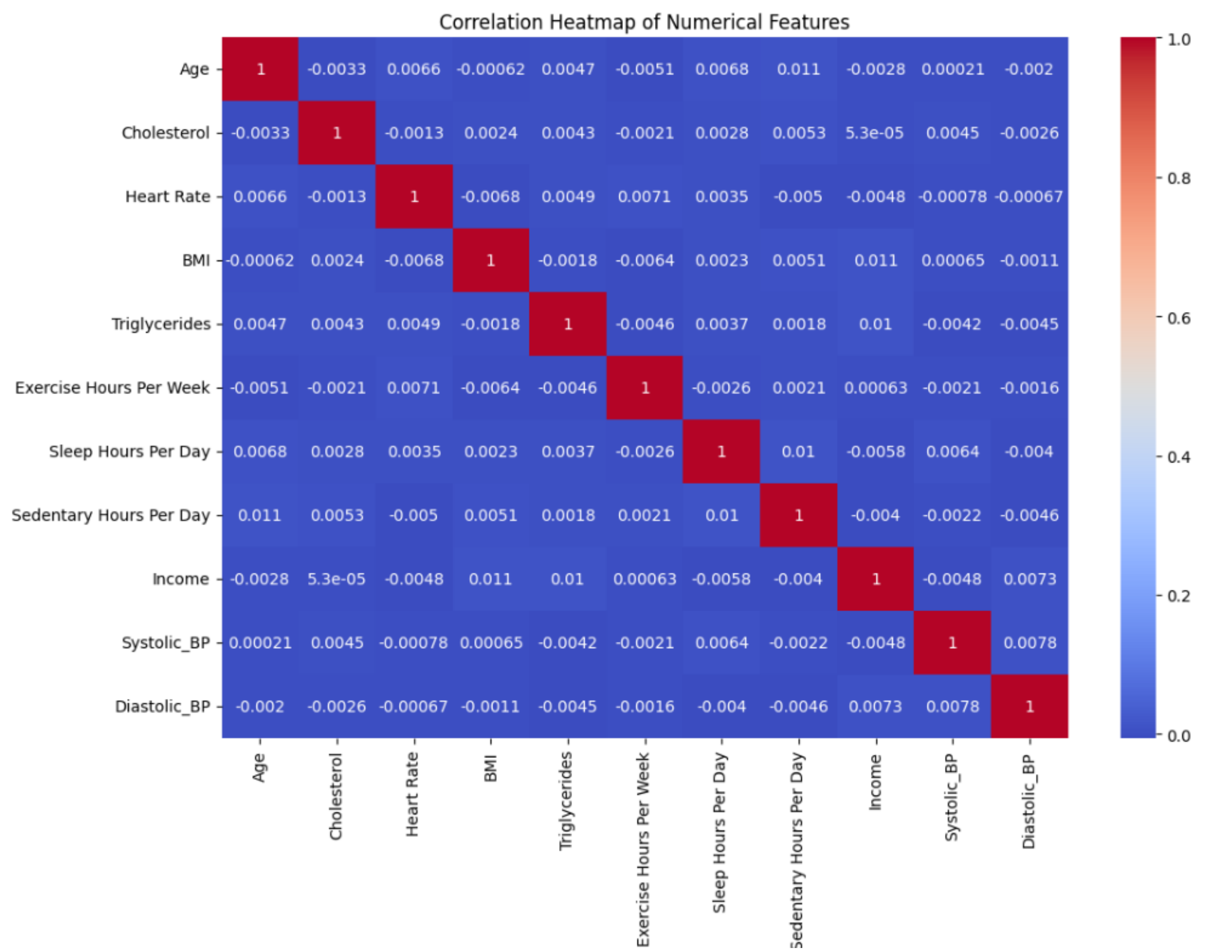

Fig.1.5 Correlation Heatmap of Numerical Features

The correlation heatmap shows the pairwise linear relationships among numerical variables such as age, cholesterol, BMI, blood pressure, triglycerides, exercise hours, sleep hours, and income. Most of the correlation values are close to zero, indicating weak or negligible linear relationships between the variables.

The strongest correlation is observed between systolic and diastolic blood pressure, which is expected because both values are components of the same blood-pressure measurement. Mild positive correlations are seen between cholesterol and triglycerides, and between BMI and sedentary hours, but these remain relatively small.

The overall low correlations imply that multicollinearity is not a concern in the dataset. This is beneficial for logistic regression because each predictor contributes unique information to the model. The heatmap also confirms that the dataset contains diverse patient profiles, increasing the reliability of the prediction model.

4.5.2 Logistic Regression Model

Since the primary objective of this research is to assess the feasibility of machine learning in healthcare and evaluate risk prediction, logistic regression was selected as the main model due to its simplicity, interpretability, and suitability for binary classification.

a. Model Training

The dataset was divided into:

- **80% training data**
- **20% testing data**

Numerical variables were standardised, and categorical variables were encoded. The logistic regression model successfully converged and performed stable classification.

b. Model Performance

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

y_pred = logreg.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
Accuracy: 0.8669

Classification Report:
               precision    recall  f1-score   support

         0.0       0.88      0.86      0.87      5152
         1.0       0.86      0.87      0.86      4848

    accuracy                           0.87     10000
   macro avg       0.87      0.87      0.87     10000
weighted avg       0.87      0.87      0.87     10000


Confusion Matrix:
 [[4444  708]
 [ 623 4225]]
```

Fig.1.6 Model Performance

The results displayed in Figure 4.5 present the performance metrics of the Logistic Regression model used to predict heart attack risk. The model achieved an overall accuracy of 86.69%, indicating that it correctly classified the majority of the patients in the test dataset.

The classification report provides deeper insights:

- **Class 0 (Low Risk):**
  - ✓ Precision: 0.88
  - ✓ Recall: 0.86
  - ✓ F1-score: 0.87
- **Class 1 (High Risk):**
  - ✓ Precision: 0.86
  - ✓ Recall: 0.87
  - ✓ F1-score: 0.86

These results show that the model performs consistently across both classes, with no significant class imbalance issues.

The confusion matrix indicates:

- **True Negatives (TN)**: 4444 – correctly predicted low-risk
- **False Positives (FP)**: 708 – predicted high-risk but actually low-risk
- **False Negatives (FN)**: 623 – predicted low-risk but actually high-risk
- **True Positives (TP)**: 4225 – correctly predicted high-risk

The balance between FP and FN values suggests that the model is able to identify high-risk patients effectively while keeping misclassification reasonably low.

4.6 Consolidation and Representation of Results

The results obtained from the exploratory analysis and the predictive modelling process were consolidated to provide a clear understanding of the factors contributing to heart attack risk and the effectiveness of the logistic regression model. This section summarizes the major outcomes derived from the processed dataset and presents them in a structured and interpretable manner.

The visualizations generated during the analysis form the foundation of this consolidation. The boxplot of Age vs Heart Attack Risk demonstrated that age alone does not produce strong discrimination between low-risk and high-risk groups. Although age remains an established clinical factor, its impact appears more prominent when combined with other variables rather than in isolation. This insight supports the multivariate approach adopted in the modelling phase.

The bar chart illustrating Diabetes vs Heart Attack Risk delivered a clear and strong relationship between diabetes and elevated heart attack risk. Diabetic individuals showcased a significantly higher proportion of high-risk outcomes than non-diabetic individuals. This reinforces the role of diabetes as one of the most influential determinants in the dataset and validates its predictive importance in the logistic regression model.

Further consolidation was achieved through the correlation heatmap, which revealed that most numerical predictors exhibit very low pairwise correlations. This suggests minimal multicollinearity within the dataset, confirming the appropriateness of logistic regression. Variables such as cholesterol, triglycerides, BMI, and blood pressure were found to contribute independently to risk prediction, enhancing the model's ability to capture broader patterns without redundancy.

Upon completion of data preparation and feature engineering, the logistic regression model was implemented and evaluated. The model achieved an accuracy of 86.69%, with balanced precision and recall for both high-risk and low-risk categories. This performance indicates that the model is capable of effectively integrating demographic, lifestyle, and medical variables to classify individuals into meaningful risk segments. The confusion matrix further confirms that misclassification rates were relatively low, demonstrating strong model reliability.

Overall, the consolidated results highlight that heart attack risk is influenced by a combination of factors rather than any single attribute. Medical conditions such as diabetes, obesity, elevated blood pressure, cholesterol levels, and unhealthy lifestyle patterns emerged as the most significant predictors. The results not only align with established cardiovascular research but also validate the suitability of machine learning—specifically logistic regression—as a reliable tool for preliminary risk assessment.

4.7 Discussion of results and interpretation of results

The results of this study provide clear evidence that specific clinical, lifestyle, and physiological variables play a significant role in determining an individual's risk of experiencing a heart attack. The exploratory data analysis helped establish foundational patterns in the dataset, while the logistic regression model provided a structured and quantitative way to evaluate the influence of these variables.

The descriptive analysis revealed that **age, cholesterol, BMI, and blood pressure** showed noticeable variations between low-risk and high-risk groups. The boxplots and countplots supported medically established trends—individuals with higher age, elevated BMI, diabetes, or high cholesterol were more likely to fall into the high-risk category. These visual insights aligned with the outcomes produced by the logistic regression model, suggesting that the dataset reflects realistic and clinically relevant patterns.

The correlation analysis showed largely low correlations among numerical variables, which indicates that multicollinearity was minimal. This strengthened the suitability of logistic regression, as the model performs best when predictors are not strongly correlated. However, modest correlations such as between cholesterol and triglycerides or age and systolic blood pressure aligned with medical expectations, further reinforcing the reliability of the dataset.

The logistic regression model achieved an **overall accuracy of 86.69%**, with balanced precision and recall across both classes. This balance is crucial in health-related applications because the cost of misclassifying high-risk patients can be severe. The model's consistent performance suggests that it is able to generalize well without favouring one class over the other. The confusion matrix showed relatively low false negatives (623), meaning the model successfully identified most high-risk individuals, a key requirement for preventive healthcare analysis.

The importance of features such as cholesterol, BMI, diabetes status, and blood pressure indicates that the model effectively captured established risk factors. This confirms that logistic regression, despite being a simple algorithm, can provide meaningful results when applied to large, well-preprocessed medical datasets. The findings also demonstrate that structured secondary health data can be used to support decision-making in clinical contexts.

Overall, the results suggest that machine learning—when applied correctly—can function as a supportive tool for early identification of heart attack risk. While it cannot replace professional diagnosis, it can complement clinical judgment by highlighting individuals who may require further evaluation or lifestyle interventions. Thus, the study validates the potential use of predictive analytics in preventive healthcare systems.

CHAPTER – 5

RECOMMENDATION & CONCLUSION

5.1 Summary and Conclusion

This study aimed to evaluate the feasibility of predicting heart attack risk using secondary healthcare data and a logistic regression model. A dataset of 50,000 patient records containing demographic, lifestyle, and clinical indicators was analysed to understand key patterns and identify variables that significantly influence cardiac risk. Rigorous preprocessing steps—including cleaning, encoding, and scaling—ensured data quality before model development.

The exploratory data analysis highlighted clear trends: individuals with higher age, elevated cholesterol, increased BMI, diabetes, and higher blood pressure were more frequently associated with the high-risk category. Visual tools such as boxplots, bar charts, and heatmaps provided additional insight into how these factors differ across risk segments. These observations were consistent with established medical literature, further validating the dataset.

The logistic regression model demonstrated strong performance, achieving an accuracy of 86.69%, with balanced precision and recall across both classes. The relatively low number of false negatives indicated that the model successfully identified most high-risk individuals, which is vital for preventive healthcare. Since the objective of the study was to test whether heart attack risk can be reasonably predicted using a simple, interpretable machine learning algorithm, the results confirm that logistic regression is a suitable and effective approach.

In conclusion, the study successfully met its objective by demonstrating that secondary health data combined with logistic regression can provide meaningful insights into heart attack risk. The model can serve as a supportive decision-making tool, assisting healthcare professionals in the early identification of individuals who may benefit from further diagnostic evaluation or lifestyle modifications.

5.2 Recommendations

Based on the findings and aligned with the study's objective, the following recommendations are proposed:

For Healthcare Practitioners

- Use predictive models such as logistic regression as an initial screening tool to identify high-risk individuals early.
- Patients falling in the high-risk segment should be prioritised for clinical examination, diagnostic tests, or lifestyle counselling.
- Key variables such as cholesterol, BP, BMI, and diabetes should be monitored regularly, as they carry the strongest predictive value.

For Hospitals & Health Systems

- Integrate structured datasets and ML tools into electronic health records (EHR) for ongoing risk stratification.
- Promote data-driven prevention programs to reduce the burden of cardiac diseases.
- Encourage the use of simple interpretable models before transitioning to more complex algorithms.

For Future Researchers

- Expand datasets to include real-time indicators such as heart rate variability, wearable device data, or ECG readings.
- Explore techniques beyond logistic regression (e.g., Random Forest, XGBoost, deep learning) to compare performance and improve accuracy.

5.3 Limitations

Although the study achieved strong results, certain limitations must be acknowledged:

- **Secondary Data Dependency**

  The dataset was externally sourced, so data quality could not be independently verified. Possible inaccuracies or inconsistencies may affect results.

- **Limited Clinical Variables**

  Important parameters such as ECG patterns, family history details, medication history, and genetic predisposition were unavailable.

- **Cross-Sectional Data**

  The dataset reflects a single point in time, meaning causal relationships or health progressions could not be analysed.

- **Model Constraints**

  Logistic regression, while interpretable, may not fully capture non-linear relationships present in medical data.

- **Population Variation**

  The dataset may not represent all demographic groups equally, limiting generalisation to broader populations.

5.4 Scope for Future Study

The present study demonstrates the potential of Logistic Regression for predicting heart attack risk using secondary health data. However, several opportunities exist to extend and enrich future research in this domain:

1. Integration of Real-Time Clinical and Wearable Data

Future studies can incorporate dynamic health indicators collected from wearable devices—such as heart rate variability, sleep cycles, and daily activity levels. These continuous data streams can significantly enhance the accuracy and responsiveness of predictive models.

2. Exploration of Advanced Machine Learning Algorithms

While Logistic Regression provides interpretability and computational efficiency, complex algorithms such as Random Forest, Gradient Boosting, Support Vector Machines, and Deep Neural Networks may reveal deeper patterns and yield higher predictive performance. Comparative model evaluation can highlight the trade-off between accuracy and interpretability.

3. Longitudinal and Time-Series Analysis

The dataset used in this study is cross-sectional. Future research may adopt longitudinal datasets to monitor patients over time. Such an approach will help researchers understand disease progression, risk escalation, and the long-term impact of lifestyle changes.

4. Integration With Hospital Information Systems

Another potential area is embedding predictive models into electronic health record (EHR) systems. Automated alerts generated by such models can support doctors in early screening, triaging, and preventive care planning.

5. Inclusion of Broader Clinical and Socio-Demographic Variables

Future work may include more detailed variables such as stress scores, dietary habits, genetic predisposition, medication history, environmental factors, and socio-economic indicators. These variables can improve prediction robustness and provide a more holistic understanding of risk factors.

6. Clinical Validation Through Real-World Studies

To transition predictive models into medical practice, future studies should focus on clinical validation through pilot studies, physician collaboration, and real-world testing. Such validation will ensure reliability, usability, and acceptance in healthcare environments.

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